NSF FUTURE MANUFACTURING DATA CHALLENGE

Predicting local DED track geometry from thermal history

Offline completed-sequence hierarchy, physically consistent boundaries, causal ablation, and nested four-track validation

0.148 mm	20.7%	0.142 mm	93.5%
four-track width MAE	lower than direct Ridge	mean boundary MAE	conditional coverage

Executive summary

A laser track is a spatial signal, not one average width. We align each active thermal frame to physical x, extract the local track center and connected left/right profilometry boundaries, and evaluate every track as an untouched condition. The promoted nested selector reduces track-balanced width MAE from 0.187 mm to 0.148 mm (20.7%), worst-track MAE by 34.8%, and boundary MAE by 21.1%.

The primary model predicts condition-level baseline center and log-width from completed-sequence thermal summaries, then predicts local residuals from frame history. It runs after the thermal scan completes and directly addresses cross-power mean shifts and over-smoothed local signals.

Problem formulation and data alignment

For each valid x location, targets are width, center, left boundary, and right boundary. Profilometry cross-sections are robustly detrended; connected 30%-of-peak crossings around the bead define the boundaries. Thermal frames map to x using 10 mm/s at 50 fps (0.2 mm/frame). Frames farther than 0.10 mm from valid geometry are excluded, and only the 24-96 mm steady-state region is scored.

Leakage controls

- Outer leave-one-track-out tests hold out Tracks 8, 10, 14, and 21 in turn.
- Feature family, estimator, preprocessing, and interval calibration are chosen inside the other three tracks by inner leave-one-track-out validation.
- The primary model may use later thermal frames from the completed scan, but no held-out geometry, test-track labels, or post-process SEM enters prediction, feature/model selection, preprocessing, or interval calibration.
- A separate current-and-past-only ablation proves that changing future thermal frames cannot change an earlier causal-ablation prediction.

Generative AI use

Generative AI assisted with code review, test generation, debugging, and document layout. Team members reviewed the resulting code and materials. All numbers come from tracked analysis code and saved outer-fold predictions; AI did not supply or alter experimental measurements.

Method: preserve condition shifts and local variation

Thermal descriptors. Instantaneous features cover hot-region area and axes, temperature percentiles, thermal mass, gradients, asymmetry, cooling-tail area/decay, centroid velocity, and pool-shape change. Causal 5-, 10-, and 20-frame windows add means, slopes, changes, persistence above fixed temperature thresholds, and history-availability flags. Robust within-track normalization isolates local deviations without exposing held-out geometry.

Constrained multi-output prediction. Completed-sequence thermal summaries predict baseline center and baseline log-width. Local features jointly predict center and log-width residuals. Exponentiating log-width guarantees positive width; left/right are reconstructed from one shared center and width, so boundaries remain ordered.

Low-capacity model ladder. Inner folds compare Ridge, elastic net, partial least squares, spline-Ridge, and a Gaussian process. Within 0.01 mm of the best inner MAE, selection favors lower variation-scale error and higher residual correlation. Each outer fold independently selects its feature family and estimator using only the other three tracks.

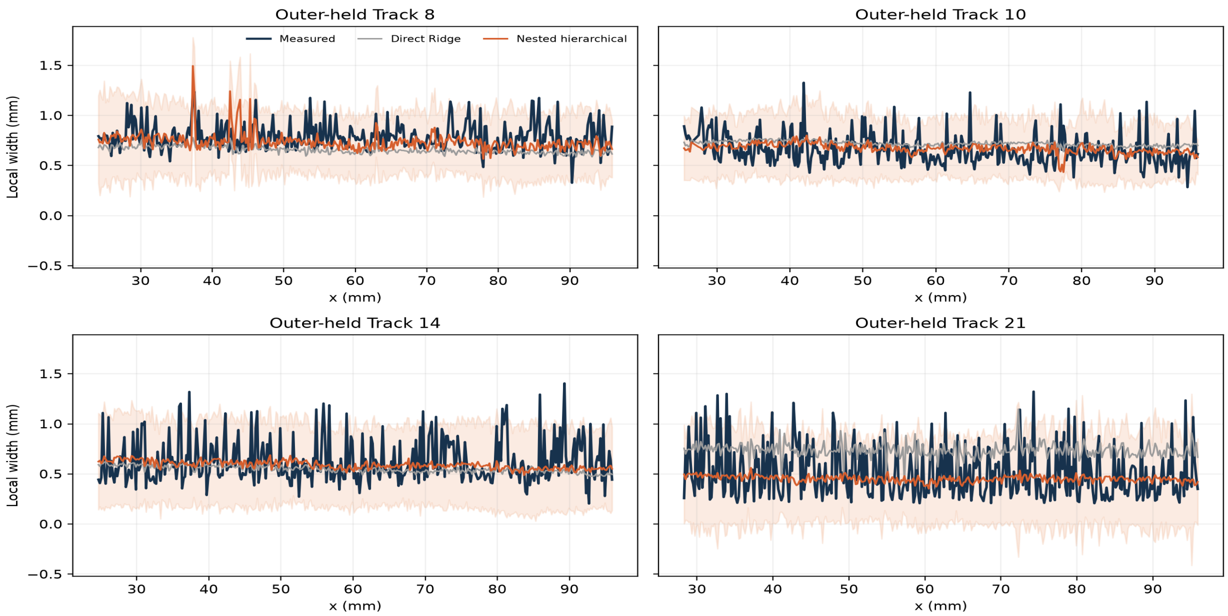

Figure 1. Measured and predicted local width for four untouched outer tracks. Each panel is produced by a model selected without that track's labels.

Metric	Direct Ridge	Hierarchical selector	Change
Track-balanced width MAE	0.187 mm	0.148 mm	-20.7%
Worst-track width MAE	0.308 mm	0.201 mm	-34.8%
Mean left/right boundary MAE	0.180 mm	0.142 mm	-21.1%
Residual correlation	0.055	0.086	1.57x
Predicted/measured local width std.	0.214	0.323	+0.109

Uncertainty, interpretation, and limits

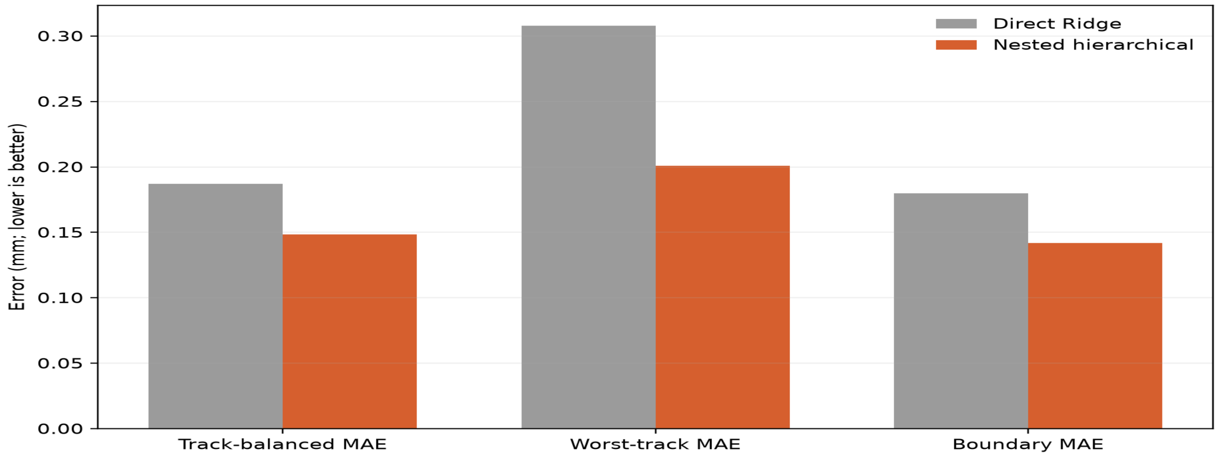

Figure 2. The promoted model improves accuracy, boundary position, and predicted variation amplitude under the same four-track outer protocol.

Conditional uncertainty

Normalized conformal intervals scale by predicted local difficulty. Conditional coverage is 93.5% with 0.780 mm mean width, versus 94.9% and 0.877 mm for a fixed global interval. The conditional method is selected because it is closer to the 90% target and narrower. Its mean width expands from 0.629 mm in easy regions to 0.944 mm in difficult regions.

Interpretable links and substrate evidence

Track-level hot area, maximum temperature, thermal mass, and cooling-tail summaries anchor the condition baseline. Normalized pool shape, temperature, asymmetry, and recent history contribute local residual information. These are predictive associations. The available SEM is post-process and is excluded from prediction because it is unavailable at frame-time inference. We therefore make no causal pre-process substrate claim.

Limitations

- Track-balanced R-squared remains -0.13; the result is not ready for closed-loop control.
- The primary result is offline. The current-and-past-only ablation scores 0.157 mm MAE and -0.18 R-squared.
- Only four independent tracks are available, so model ranking and interval coverage remain uncertain across new plates and recipes.
- The earlier 0.139 mm Track 21 result is a historical tuned split. Under the stronger nested protocol, Track 21 MAE is 0.201 mm.
- Genuine pre-process SEM or surface measurements registered to physical x are needed to distinguish substrate-driven from process-driven variation.

Conclusion

Separating condition-level geometry from local residual variation produces a more accurate, less over-smoothed, physically consistent predictor. The next decisive study should add powers, plates, repeats, and registered pre-process surface measurements, then test the locked pipeline on blind tracks.

Reproduction and sources

Reproduce with scripts/run_improvement_experiments.py. Dataset: doi:10.5281/zenodo.21285367; paper: arXiv:2607.07965; metrics: results/improved_submission/metrics.json.